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Real-time molecular specific BTEX detection at TLV-TWA concentration levels

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ABSTRACT

Volatile Organic Compounds (VOCs) exhibit a wide range of toxicity, depending on the molecular nature of specific VOCs and their associated concentrations. Existing sensing technologies are unable to determine the molecular nature of VOCs in real time, rendering the determination of a sample containing a complex VOC mixture as toxicologically safe or hazardous impossible. Herein, we demonstrate the molecular specific, real-time detection of VOCs derived from benzene, toluene, ethylbenzene, and ortho and para xylene isomers — jointly referred to as BTEX gases — within a gas mixture using a technique coined as Sample Ionisation Potential Spectroscopy (SIPS). Our solution is based on photoionization detection, utilizing tuneable, quasi-monochromatic vacuum ultraviolet (VUV) light generated by third-harmonic generation (THG) in suspended dielectric nanomembranes. The results demonstrate the potential of SIPS to identify individual BTEX compounds based on their ionisation potentials within a background matrix of five BTEX compounds in nitrogen, with a sensitivity limit down to concentration levels corresponding to occupational exposure limits.

1. Introduction

A lack of clean air represents one of the greatest environmental threats to human health and has been acknowledged as a leading cause of non-communicable diseases such as heart attacks or stroke. According to the World Health Organization (WHO), there are seven million premature deaths every year due to the combined effects of outdoor and household air pollution– with millions more people falling ill from breathing polluted air. Particulate Matter (PM) and Volatile Organic Compounds (VOCs) are the prime constituents of air pollution. Toxicologists have widely investigated the adverse effects of air pollution on human health, reporting substantial societal and economic impacts (Brauer et al., 2012; Burnett et al., 2014; Zhou et al., 2023; Technical note_Estimating the external costs of industrial air pollution-Trends, 2012-20).

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Volatile organic compounds (VOCs) are defined as organic compounds that have an initial boiling point less than or equal to 250 °C measured at a standard pressure of 101.3 kPa (European Parliament and Council, 2004), implying that they are compounds that easily evaporate under normal indoor atmospheric conditions of temperature and pressure. VOCs are known to originate from anthropogenic or biogenic sources. (Park et al., 2013) Identification and characterization of VOCs benefit economic areas like indoor air quality, industrial pollution, food, agriculture and health monitoring.

Commercial VOCs sensors can be classified based on their physical measurement principle as metal oxide (MOX) (Seiyama et al., 1962), non-dispersive infrared (NDIR) (Burough, 1980; Sun et al., 2023) and photoionisation detectors (PID) (Miranda and De Beule, 2021; Wang et al., 2023; PID Handbook). While MOX and NDIR sensors are primarily utilized to detect low molecular weight VOCs such as CO, CO_2, and CH_4, PIDs can only be applied for the detection of higher molecular weight VOCs such as benzene for example (Vilarinho et al., 2025). PID sensors can collect data in real-time, but they lack molecular selectivity reporting a total VOC concentration (TVOC) (El-Hashemy and Ali, 2018). The only known technique to selectively detect high molecular weight VOCs is Gas Chromatography-Mass Spectrometry (GC−MS) that has been the gold standard technique when molecular selectivity is required. However, GC−MS is a laboratory-based technique requiring sampling, preconcentration and ultra-high vacuum and is not in real-time.

Chemicals from the BTEX (benzene, toluene, ethylbenzene, and xylene isomers) group have been widely identified and characterised as air pollutants in outdoor and indoor environments (Du et al., 2014; Filippini et al., 2019; Kreis et al., 2022; Shojaei and Rostami, 2024; Zahed et al., 2024). While in the outdoors they derive primarily from fossil combustion, gasoline, vehicle exhaust, their presence indoors is attributed to, e.g. furniture and construction materials (Du et al., 2014). Benzene, toluene and xylene isomers are also known as ozone precursors and their interaction has implications for climate effects and chemistry in the atmosphere (Pernov et al., 2021). From a toxicological perspective they are associated with different health hazards: benzene is classified as human carcinogen (Agency for Toxic Substances and Disease Registry (ATSDR), 2017; Wang et al., 2024); toluene linked to brain and nervous system effects (Agency for Toxic Substances and Disease Registry (ATSDR), 2017); ethylbenzene is an animal carcinogen (Agency for Toxic Substances and Disease Registry (ATSDR), 2010); and xylenes have been reported to cause respiratory irritation (Agency for Toxic Substances and Disease Registry (ATSDR), 2007). Aside from links between BTEX and incidences of overall and site-specific cancers (Yu et al., 2025), BTEX is recognized as an inhibitor of cognitive development of schoolchildren (Lucialli et al., 2020).

Dedicated BTEX monitors have been commercialized based on PID technology with a gas input modification including a micro-electro-mechanical-system preconcentrator and miniature GC step (Aeroqual, 2025a). According to system specifications (Aeroqual, 2025b), a read-out of BTEX concentration every 15 min with an operational concentration range of 0.1–50 ppb is feasible. The latter concentration range is unfortunately not compatible with monitoring the trespassing of threshold limit value - time weighted average (TLV-TWA) concentration levels, considered a threshold for occupational exposure. One further restriction of miniature GC system is the reduced column length (Hsieh and Kim, 2016) rendering the elution peaks of the various compounds very broad compared to reference GC−MS systems. The latter severely restricts the robustness of said BTEX monitors against false detection induced by confounding compounds that will appear in the same elution time window. In summary, with the present state-of-the-art of gas sensors toxicologists experience a key limitation in evaluating air quality. They are namely not capable of reporting real-time information on the total VOC content in the air distinguishing benign from toxic compounds. If a GC-MS study is utilised, a qualitative breakdown of all VOCs present in the air can be provided through a time and labour-intensive laboratory study unable to capture critical time dependent concentration fluctuations that can be blinded from long term passive or active sample collection. Hence, it is of paramount importance to develop an innovative solution that can retrieve molecular selectivity and real-time detection at atmospheric pressure.

Beyond the commercial BTEX monitoring solutions, several new avenues of detection are currently being explored. Non-selective BTEX gas detection systems have been presented based on electrochemical gas sensors (Sekhar and Subramaniyam, 2013) and UltraViolet-Visible absorption spectroscopy (Allouch et al., 2013). Using mid-infrared spectroscopy at 3.3 μm, deep neural networks have been applied for the real-time selective detection of BTEX (Mhanna et al., 2022). A strong overlap of the absorption spectra of hydrocarbons at 3.3 μm limits the capacity to discriminate molecular compounds. At longer light wavelengths 13–15 μm, BTEX absorption spectroscopy show more distinct features. This has been explored through the application of InAs-based quantum cascade lasers specifically designed to target benzene and toluene spectral absorption profiles and a quartz tuning fork for enhanced photoacoustic spectroscopy (Ma et al., 2018). This approach faces the challenge whereby a different laser source is required for the targeting of each compound of interest and the presence of considerable side-lobes in the absorption spectra of many compounds, including CO_2 (Karhu et al., 2020), limits the capacity to act as a highly selective BTEX sensor in a real-world contaminated environment. At the other side of the electromagnetic spectrum, a tuneable light source in operating in the middle ultraviolet (200 – 300 nm) has been utilized to detect BTEX compounds with multiphoton electron extraction spectroscopy (MEES). This approach demonstrated ultrasensitive detection in the parts per trillion range of BTEX compounds and demonstrated spectral separation of a toluene/benzene mixture through a spectral wavelength scan between 257.5 nm and 262 nm at a resolution of 0.02 nm.

Here, we present a novel VOC gas sensor design for photoionization entirely under atmospheric conditions based on quasi-monochromatic tuneable wavelength vacuum ultraviolet (VUV) (2.1.). Subsequently, we define a new data acquisition protocol referred to as Sample Ionisation Potential Spectroscopy (SIPS), capable of molecular specific, real-time detection of VOCs (2.2.). Timing and flow conditions for an experimental SIPS data acquisition protocol are defined (2.3.), capable of demonstrating that the signals detected can be attributed to individual BTEX compounds within a BTEX mixture (2.4.). In section 3 we demonstrate the molecular specific detection of individual BTEX compounds within a BTEX mixture occupational exposure limit, equivalent to TLV-TWA mentioned above, established by the European Union. This achievement represents a significant step forward in the road towards real-time toxicological evaluation of air quality hampered by the presence of a complex mixture of VOCs, without the use of laboratory grade reference GC-MS instrumentation.

2. Experimental methodology

2.1. Photoionization based VOC sensor based on tuneable quasi-monochromatic VUV light

A traditional PID sensor for VOC detection consists of a low-pressure excitation lamp-based VUV light source (Fig. 1A). VUV light is exposed to an air sample containing VOCs within an ionisation volume (Fig. 1B). When the VUV light photons carry sufficient energy ($E = hc/\lambda$) to knock off an electron from chemical compounds within the air, the generated ions are translated by an electric field to a current detection electrode, generating a very small current in the process that is typically within the pico-to nanoampere range. In a last step depicted in Fig. 1C, the generated current is amplified by a transimpedance amplifier to a voltage signal, bearing a direct relationship to the molecular concentration of the VOCs that were ionised. The introduction of a VUV light source that can be tuned in the wavelength range allows for compound screening based on the molecular first ionisation potential, which can be considered as a selective marker for a specific VOC within a broad background.

We have previously introduced a novel VUV light source for photoionization (Vilarinho et al., 2025) and applied it for the linear detection of ethanolamine and eucalyptol in the gas phase for concentrations near the saturated vapour pressure (SVP). A high-power ultrafast pump laser operating at 1030 nm (Carbide CB5, Light Conversion, Lithuania) provides the pump source for an Optical Parametric Amplifier (OPA) delivering a femtosecond (~230 fs) pulsed (60 kHz) light source in the wavelength range of 720 nm to 880 nm (Orpheus, Light Conversion, Lithuania). This light is subsequently downconverter to the range of 360 nm to 440 nm using the Second Harmonic Generation (SHG) process in a BBO crystal (Lyra, Light Conversion, Lithuania). Down conversion to the VUV regime (120 nm to 180 nm) is finally achieved utilizing suspended SiO_2 dielectric nanomembranes (NMBs).

The photoionization-based sensor applied in this work represents a second-generation integration of the sensor as depicted in Appendix A. A new ionisation chamber was designed and built in-house, consisting of anodized aluminium to reduce potential VOC background through surface contamination, with a significantly reduced dimension of 100 mm × 38 mm × 18 mm (height x length x width), providing an airtight volume with gas inlet and outlet, creating an ionisation chamber volume of 307 $mm^3 \approx 3.10^{-4}$ l. This implies that considering a flow rate of 200 mL/min, the full ionization chamber gets renewed in ~6 s. The thickness of the SiO_2 suspended nanomembranes has been reduced from 600 nm to 300 nm, strongly increasing the efficiency of the THG process creating the VUV light (Vilarinho et al., 2025). This reduction has been achieved by a modified cleanroom process that reduced the diameter of the nanomembranes to 0.5 mm. To reduce the confounding impact of electronic noise, the suspended dielectric nanomembranes have been integrated onto a printed circuit board hosting an integrating transimpedance amplifier as depicted in Appendix A.1. The laser beam is focused to a spot < 100 μm by a CaF_2 focusing lens on one nanomembrane of the array at a time. An electric field has been applied to our nanomembranes device by connecting 50 V DC voltage to the pads on top of the device using a high voltage power supply (Stanford Research Instruments Inc., Model PS 350/5000 V−25 W), creating a field strength of ~100 V/cm^{-1} given that the pads have a separation of ~5 mm, in-line with field strengths applied in Ion Mobility Spectrometry (Eiceman et al., 2013). The generated current in the picoampere to nanoampere range is integrated for a determined period (200 ms) and amplified by an integrating transimpedance circuit to transform it to a detectable voltage. This voltage signal is then recorded by a data acquisition system (Digital I/O module 782608 −01, National Instruments) and sent to the PC for storage and post-processing using a dedicated software written in LabVIEW 2023. Once molecules of a gas matrix confined in the ionisation chamber are exposed to the selected THG of the pump wavelength originated in the dielectric nanomembranes the photoionisation process occurs. The metallic electrodes deposited generate an electric field that collects the VOC ions created. The transimpedance amplifier records the current through voltage detection with an external PC equipped with voltage detection circuitry.

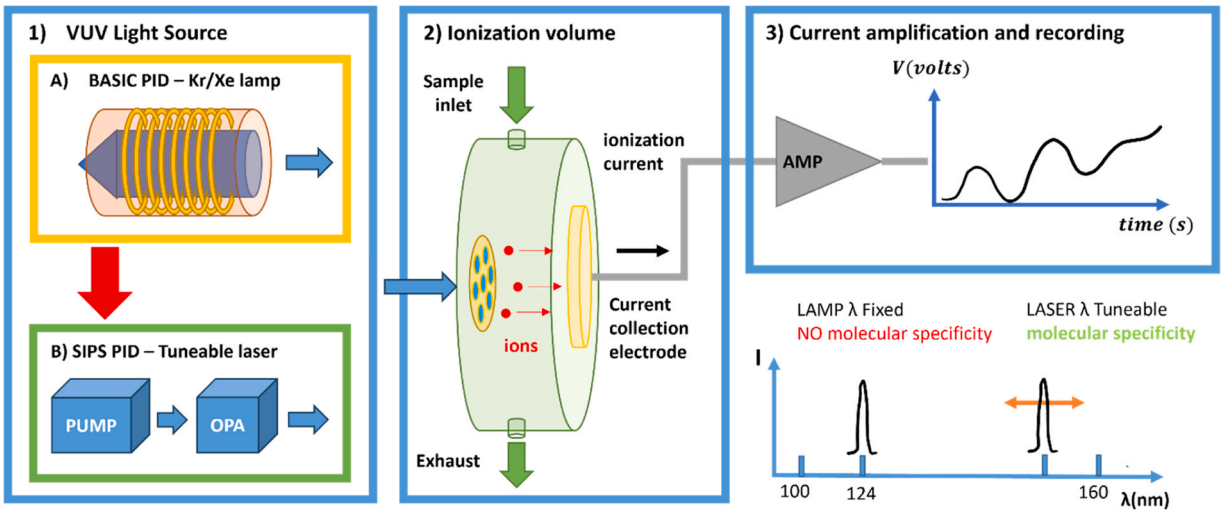


Fig. 1. Traditional lamp based PID versus SIPS based PID.

2.2. SIPS data analysis

Each chemical compound consisting of atom or molecule in the gas phase has a characteristic ionisation potential (IP), ie., a minimum energy necessary to eject an electron. To induce ionisation with light, a process referred to as photoionization, the photons composing the light field need to possess an associated energy higher than the IP of the chemical compound. For almost all chemical compounds the latter implies that the light inducing photoionization needs to have an associated wavelength (λ_{ph}), inversely proportional to the photon energy (E_{ph}), in the vacuum ultraviolet (VUV) wavelength range between 100 nm to 200 nm.

Fig. 2 details how a quasi-monochromatic VUV light source with continuous tuning capability can be utilised to derive concentration levels of specific chemical compounds within a complex mixture through photoionisation. Most VOCs exhibit an ionisation potential in the range of 7.75 eV to 11.80 eV, corresponding to a VUV wavelength range of 160 nm to 105 nm respectively. The IP of a VOC represent a threshold by which photoionisation occurs when the photon energy is higher than the IP. Hence, the total photoionization current response signal (I) of a complex mixture of VOC with carrying IPs will continuously rise as shown in Fig. 2b. Additionally, this implies that the concentration value of a VOC with IP E_{IP} is proportional to $\frac{\partial I}{\partial E_{ph}} r^2$ evaluated at E_{IP} as illustrated in Fig. 2c.

In this manuscript, we implement the spectroscopic approach depicted in Fig. 2c in a two-wavelength switching method for each BTEX compound in the mixture described in 2.4. with wavelengths λ_{VUV-H} and λ_{VUV-L} just below and above the wavelength associated with the ionization potential λ_{IP} respectively, approximating the derivative through

$$\frac{\partial I}{\partial E_{ph}} \cong \frac{I(\lambda_{VUV-H}) - I(\lambda_{VUV-L})}{E_{ph}(\lambda_{VUV-H}) - E_{ph}(\lambda_{VUV-L})}$$

2.3. SIPS signal data acquisition protocol

A SIPS acquisition routine was developed to fully characterize the interaction of the VUV laser light in our SIPS sensor with a BTEX matrix consisting of each BTEX compound at TLV-TWA concentration, as depicted in Fig. 3. The design of this routine enables the attribution of currents detected to a photoionization process. Since we are dealing with volumetric gas, fluxes stabilization periods are incorporated in the protocol. A nitrogen background flow of 200 mL/min was maintained through the ionization volume throughout the data acquisition protocol. Signal detection occurred at normal temperature and pressure (NTP) and applying a bias voltage of 50 V to create a driving electric field for an enhanced collection of the generated photoionisation current. After 30 s the VUV laser was switched on and after 90 s (180 s for benzene given the lower TLV-TWA concentration) the BTEX VOC matrix was inserted in the nitrogen background flow. Subsequently, the VUV laser wavelength was alternated twice between a pump wavelength respectively above (λ_{VUV-H}) and below (λ_{VUV-L}) the wavelength derived from the ionisation potential (λ_{IP}) of a given compound while the gas matrix flowing into the ionisation chamber. The laser was switched off at 570 s (1140 s for benzene).

2.4. Chemical compounds

A gas cylinder containing a calibrated mixture of benzene (C_6H_6, 0.5 ppm), toluene ($C_6H_5CH_3$, 50 ppm), ethylbenzene (C_8H_{10}, 20 ppm), para-xylene ($p-C_6H_4(CH_3)_2$, 50 ppm) and ortho-xylene ($o-C_6H_4(CH_3)_2$, 50 ppm) in nitrogen (N_2) was purchased from Linde Portugal. The ionisation potential (E_{IP}) values were obtained from the National Institute of Standards and Technology Chemistry (NIST) WebBook (Linstrom and Mallard, 2001). The concentrations of the VOCs were chosen taking as reference the occupational exposure limits (OEL) for these chemicals over a time-weighted average (TWA) 8-hour workday and 40-hour work week in the European Union as expressed in the International Chemical Safety Cards (ICSCs). For benzene, toluene and ethylbenzene that present high safety risks we set a more stringent detection limit. For benzene instead of 1 ppm we used 0.5 ppm and for ethylbenzene instead of 100 ppm we used 20 ppm. For the xylene isomers, that present lower safety risks in comparison with the previous ones, the concentration 50 ppm equals the OEL-TWA.

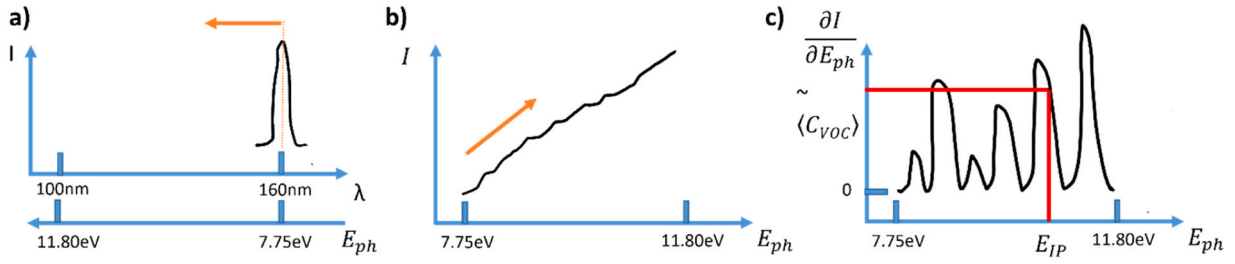


Fig. 2. SIPS theoretical analysis: a) continuously tuneable quasi-monochromatic VUV light intensity I; b) photoionisation current from a complex mixture of VOCs while reducing the light source central wavelength λ; c) concentration (C) the derivative given by $\frac{\partial I}{\partial E_{ph}}$ of a VOC with IP E_{IP} is proportional to its concentration.

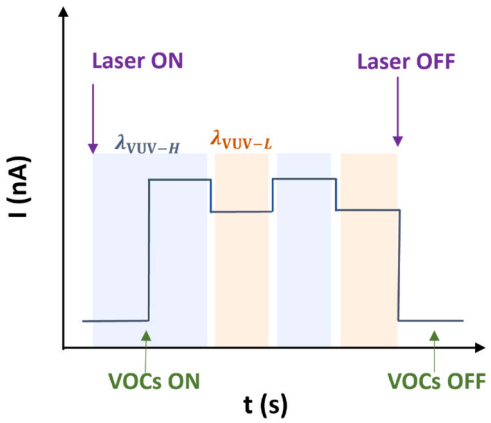


Fig. 3. Schematic drawing for the acquisition sequence routine of the photoionisation signal (I).

3. Results and discussion

Five separate measurements according to the SIPS signal data acquisition protocol detailed in 2.3 were carried out with the second-generation SIPS sensor described in 2.1 on the BTEX gas mixture in a N_2 background detailed in 2.4. The laser illumination wavelengths λ_{VUV-H} and λ_{VUV-L} applied in these five measurements were chosen to enable the detection and measurements of individual BTEX compounds within the BTEX mixture and are detailed in Table 1, referencing the reported ionisation potential of the BTEX compound expressed in both electron volts and nanometre units. The core idea behind these experiments is that if one subtracts the photoionisation current observed through SIPS at λ_{VUV-H} wavelengths from the photoionisation current at λ_{VUV-L}, one obtains the exclusive photoionisation induced by the compound for which λ_{VUV-H} and λ_{VUV-L} have been chosen, in-line with the SIPS data analysis methodology outlined in 2.2. The latter implies the sensor selectivity of a single compound within a complex mixture of five different BTEX compounds at concentrations detailed in Table 1, taking as reference the OEL-TWA established by the EU as explained in 2.4.

Fig. 4 shows the SIPS current signal acquired for the five compounds in the BTEX gas matrix with the acquisition conditions shown in Table 1. At the start of the experiment, a stable background signal of 40 nA background current is noted. Important to notice here is that switching on the SIPS laser illumination does not alter the background signal, contrary to the observations in the first generation SIPS device where an additional background signal was observed and explained through electron-hole pair formation in the SiO_2 nanomembrane (Vilarinho et al., 2025). We attribute this change to the altered photonic chip design detailed in Appendix A.1. Inducing BTEX compounds in the N_2 gas flow consistently results in a current spike beyond the background current, attributed to photoionisation. When the wavelength of the OPA is tuned for generating λ_{VUV-H} to λ_{VUV-L} via THG, a consistent decrease in current is observed, referred to as ΔI_1. Subsequently, this wavelength tuning step is repeated to evaluate measurement reproducibility, resulting in a second current decrease ΔI_2. The resulting ΔI_1 and ΔI_2 are summarized in Table 2. The response time for the current to change upon wavelength changes is on the order of seconds, confirming the real-time nature of our detection technology. An average increase of 20% is observed of ΔI_2 relative to ΔI_1, indicating that a small percentage of VOCs might get attached to the walls of the ionisation volume and tubes, resulting in a slightly higher observed signal when the experiment is repeated. The sensitivity of the SIPS sensor, as defined by the current response relative to the VOC concentration, is just as for traditional PID sensors, compound specific. Finally, the laser is turned off causing a drop in the current signal down to the original background values. This indicates that although there is a BTEX gas flux in the ionisation chamber, no photoionisation takes place.

The experimental data reported in Fig. 4 obtained for the discrete set of wavelengths defined in Table 1, can be converted into a representation like the one in 2.2, as illustrated in Fig. 5. Fig. 5a reveals the photoionization of a mixture of VOCs whereby a monotonously increasing current is observed for a rising photon energy of the light utilized to ionize the gas matrix, as depicted in Fig. 2b. The derivative of the photocurrent with respect to the photon energy (Fig. 2c) has been defined for the two-wavelength switching method as described above and represented in Fig. 5b. The application of known concentrations of BTEX compounds in a mixture as defined in 2.4 calibrates our VOC sensor quantitatively, whereby it should be stressed that each molecular species exhibits a different calibration factor. The photoionization current observed for VOC compound ionization has been determined to be linear

Table 1
BTEX properties and parameters:Ionisation Potentials, pump wavelengths and concentrations used.

VOC	E_{IP}(eV)[29]	λ_{IP}(nm)	λ_{VUV-H}(nm)	λ_{VUV-L}(nm)	Concentration (ppm)
Benzene	9.24	134.2	133.3	135.0	0.5
Toluene	8.83	140.4	139.3	140.7	50
Ethylbenzene	8.77	141.4	140.3	141.7	20
o-Xylene	8.56	144.9	144.0	145.7	50
p-Xylene	8.44	146.9	144.3	147.3	50

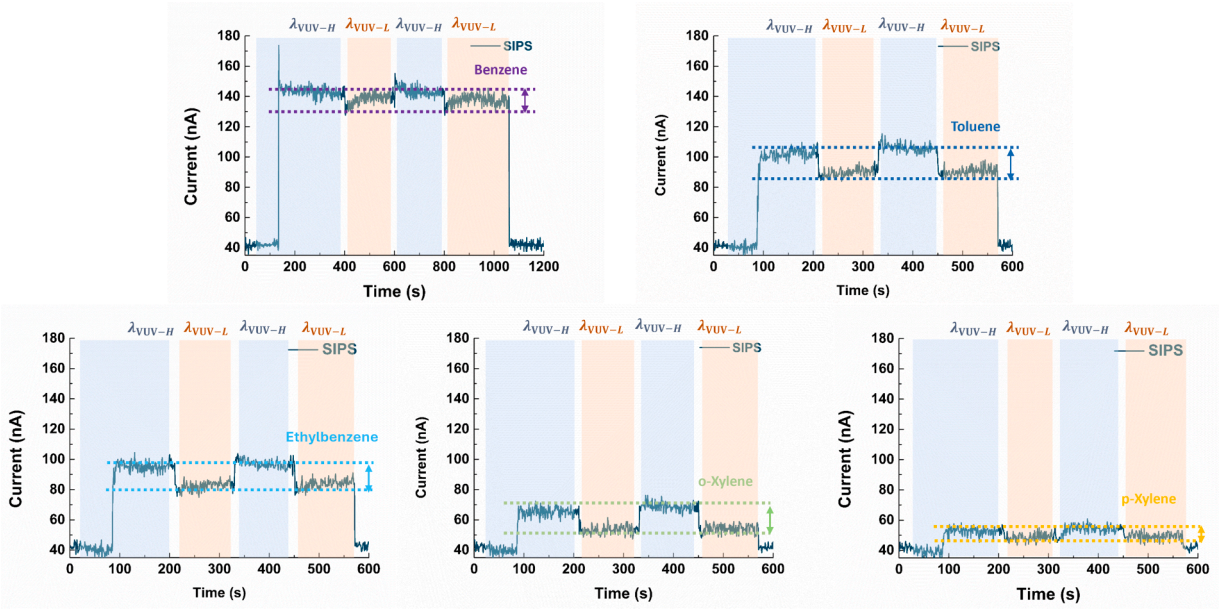


Fig. 4. SIPS response to a mixture of BTEX. A photoionisation induced current jump is observed moving the VUV wavelength from values just above λ_{VUV-H} (nm) and below λ_{VUV-L} (nm) the wavelength derived from the molecular IP for: (a) benzene, (b) toluene, (c) ethylbenzene, (d) o-xylene, and (e) p-xylene.

Table 2
SIPS photoionization signals: results obtained for each VOC.

VOC	ΔI_1 (nA)	ΔI_2 (nA)	($\Delta I_2/\Delta I_1$)	SIPS sensitivity S_{BTEX} (nA/ppm)	Limit of Detection (LOD_{VOC})) (ppm)
Benzene	5.15	6.12	1.19	10.29–12.25	0.26
Toluene	11.95	15.73	1.31	0.24–0.31	12.22
Ethylbenzene	12.77	13.44	1.05	0.64–0.67	5.13
o-Xylene	11.91	14.42	1.21	0.24–0.29	12.23
p-Xylene	4.17	5.09	1.22	0.08–0.10	25.95

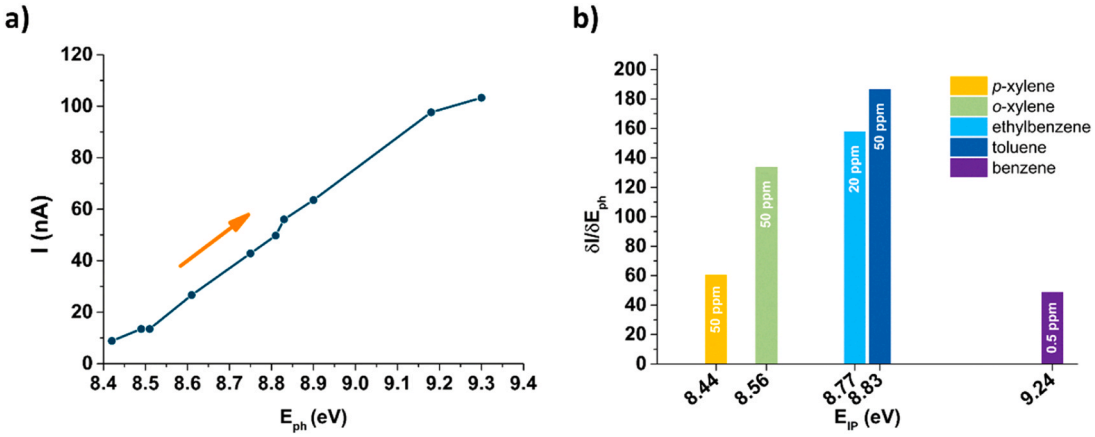


Fig. 5. (a) Sample Ionisation Potential Spectrograph of BTEX mixture in nitrogen; (b) Derivative of the photocurrent.

with respect to concentration (Vilarinho et al., 2025), implying that unknown concentrations of BTEX compounds can be defined through the calibration factors contained within Fig. 5b. System detection linearity implies that the limit of detection (LOD_{VOC}) can be defined from signal fluctuations observed under stable exposure conditions according to the equation defined as

$$\mathrm{LOD}_{VOC} = \frac{\Delta I_{noise}}{\Delta \mathrm{I}_{VOC}} C_{VOC}$$

whereby the current drop ($\Delta \mathrm{I}_{VOC}$) exerted by a known concentration of a specific VOC (C_{VOC}) is compared to the current fluctuations observed under stable operating conditions (ΔI_{noise}). The LOD for each of the BTEX compounds is different and detailed in Table 2.

Any BTEX molecular compound in concentrations beyond the defined LOD (C_{BTEX}) can be determined through the subtraction of the photoionization current observed at λ_{VUV-L} from the current at λ_{VUV-H}, referred to as $\Delta \mathrm{I}_{BTEX}$ and relating it to the SIPS sensitivity defined in Table 2 according to

$$C_{BTEX} = \frac{\Delta \mathrm{I}_{BTEX}}{S_{BTEX}}$$

C_{BTEX} can be determined a timeframe of ~1 s and is principally dependent on the integration time of the integrating transimpedance amplifier and time required to shift the wavelength emission of the OPA. Hence, we consider SIPS technology capable of real-time concentration determination.

The experiments reported in Fig. 5 have been conducted in a nitrogen atmosphere. The constituents encountered in the air atmosphere beyond nitrogen, i.e. oxygen, argon, neon, helium, methane, krypton, hydrogen, nitrous oxide and xenon cannot be ionized with the system reported here due to the extremely high ionization potentials of these constituents and the inability to induce multiphoton ionization due to limited VUV power levels generated via the THG process. This is like commercial lamp-based PID technology where the background carrier gas being nitrogen or compressed air has no influence on the VOC measurements. This contrasts with MEES technology (Sharon et al., 2025), where highly energetic light pulses of several mJ lead to a competition between three-photon ionization of oxygen and two-photon ionization of BTEX compounds. Nonetheless, choosing appropriate wavelength windows, MEES has been able to demonstrate extremely sensitive detection in the parts per trillion range of BTEX compounds as well as discriminating between toluene and benzene through a wavelength scan between 257.5 nm and 262 nm at a resolution of 0.02 nm, vastly beyond the LODs of SIPS reported here. The LOD demonstrated for SIPS in this work of ~0.3–26 ppm is encouraging, albeit real-world applications require lower LOD as many VOC species in the atmosphere are present at ppb concentration levels. We foresee two pathways to achieve this. First, a straightforward and brute force approach is to increase the laser pulse energy from the μ to the mJ range. Given that the THG process exhibits a cubic dependence on the input power (Boyd, 2008), this will vastly enhance VUV generation from the NMBs to perform photoionization. The disadvantage of this approach implies that the laser technology requires water cooling instead of the air cooling required for the laser system detailed here, implying additional complexities and an enhanced laser energy consumption hindering field deployment of the technology. A second approach would be to enhance THG efficiency in the NMBs. This can be achieved through the creation of specific nanostructures on the NMBs to generate anapoles or so-called dark modes, generating highly increased light energy in the nanostructures, leading to a substantial increase in THG efficiency (Semmlinger et al., 2019). We have recently demonstrated for the first time, the existence of anapoles excited in the UV-A region, required for THG generation in the VUV (Poeta et al., 2024). Another possible approach to increase THG efficiency is through the creation of transient Kerr gratings achieved through overlapping two equal beams in space and time at a fixed spatial angle (Punjal et al., 2026). This approach however remains to be demonstrated for the generation of VUV.

PIDs suffer from performance reduction over their operational lifetime due to surface adsorption and contamination onto the VUV lamp surfaces consisting of MgF_2 or LiF transmission windows (Herzig et al., 1992; Taylor et al., 1965). Similarly, it can be expected that SIPS will suffer from contaminations onto the SiO_2 NMBs hindering long-term operation in real-world environments. We expect that these contaminations can be mitigated using different suspended NMBs on the same chip, or through exchange of the entire silicon wafer with suspended NMBs displayed in Figure A6a. Furthermore, we expect that SIPS to exhibit a similar response to humidity as PIDs, given the similar detection methodology. Humidity is known to cause water vapor quenching reducing PID response for increased humidity. Another phenomenon caused by humidity is condensation on the sensor leading to current leakage. While the former effect is typically dealt with humidity correction tables, the latter can be avoided by warming the PID to the measurement temperature before entering a humid environment.

The SIPS measurement principle detailed above can be applied for the detection of any VOC compound within a complex background of VOCs. To determine the concentration of a VOC of interest, we first need to determine the ionization potential E_{IP} from a database reference or a quantum chemistry software package such as ORCA (https://www.faccts.de/orca/). Subsequently, we convert E_{IP} to its associated wavelength λ_{IP} and establish wavelengths λ_{VUV-H} and λ_{VUV-L} so the relationship $\lambda_{VUV-H} < \lambda_{IP} < \lambda_{VUV-L}$ is satisfied. The spectral separation between λ_{VUV-H}, λ_{IP} and λ_{VUV-L} required needs to consider the laser spectral bandwidth which is highly dependent on the laser pulse duration (Nikitin et al., 2025). Next, SIPS current measurements need to be established at λ_{VUV-H} and λ_{VUV-L} for a known concentration of the VOC compound under consideration, leading to a current change $\Delta I_{VOC-CAL}$, representing a SIPS calibration measurement for this compound. If one wants to determine the presence of the VOC within a complex background consisting of many other VOCs with different ionization potentials (E_{IP-BG}), following scenarios are possible:

- $E_{IP-BG}$is higher than the energy associated with the light at wavelength λ_{VUV-H}: the background compound will not be ionized and detected through SIPS.
- $E_{IP-BG}$is lower than the energy associated with the light at wavelength λ_{VUV-L}: the background interfering compound will be ionized both through the measurements at λ_{VUV-H} and λ_{VUV-L} : a background current I_{BG} will be observed that will be cancelled out in the concentration determination for the VOC to be monitored through SIPS:

$$\frac{\partial I}{\partial E_{ph}} \cong \frac{[I_{VOC}(\lambda_{VUV-L}) + I_{BG}] - [I_{VOC}(\lambda_{VUV-L}) + I_{BG}]}{E_{ph}(\lambda_{VUV-H}) - E_{ph}(\lambda_{VUV-L})} = \frac{I(\lambda_{VUV-H}) - I(\lambda_{VUV-L})}{E_{ph}(\lambda_{VUV-H}) - E_{ph}(\lambda_{VUV-L})}$$

- E_{IP-BG} falls between the energy associated with the light at wavelength λ_{VUV-H} and λ_{VUV-L}: in this case, the background compound will appear as a confounding compound in the SIPS measurement. Luckily, the ionization potential of many abundant chemicals exhibits a broad spectrum (https://webbook.nist.gov/chemistry/) limiting the possibility of confounding measurements.

Based on the results obtained we have compiled in Table 3 a tentative qualitative comparison between SIPS and the commercially available solutions such as PID or GC-MS technology, suitable for the detection of VOCs with a relatively high molecular weight associated with $E_{IP} < 11$eV. The fast detection and high selectivity for well-defined VOCs in real-world field environments represent a significant challenge for PID and GC-MS technology, represent an opportunity for SIPS technology. Further technology development on lowering SIPS LOD will allow for more toxicity monitoring applications to get within the scope of the technology.

4. Conclusions

The molecular specific detection of VOC pollutants is a critical aspect of air quality monitoring; gas toxicity varies significantly between individual VOCs. This is evidenced by the wide range of toxicity threshold concentration values. In this work, we demonstrate for the first time the molecular specific and real-time detection of a calibrated mixture of BTEX VOCs using a novel technique coined as SIPS. Unlike conventional PID sensors, reporting TVOC, and portable GC−MS (not in real-time), SIPS approach offers real-time and molecular selective detection. We show that SIPS can selectively detect BTEX compounds at threshold limit value–time-weighted average concentration levels in real time, without the use of bottled carrier gases and with immunity to confounding background signals from other VOCs. Since SIPS operates entirely under atmospheric pressure conditions, future field testing will be feasible. Furthermore, the SIPS sensor can be tuned to detect any VOC compound by adjusting the VUV wavelength to match the ionisation potential of the target VOC. Therefore, it is not limited to BTEX detection, paving the way for real-time assessment of air toxicity. We believe this work represents a significant advancement in the selective, real-time monitoring of pollution caused by high molecular weight VOCs, which have traditionally been assessed using PID technology. Further work will focus on real-world validation of SIPS in polluted environments, including detailed response curves to humidity, temperature and pressure. Also, several approached towards improving the LOD down to the ppb range will be explored.

CRediT authorship contribution statement

Adelaide Miranda: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Funding acquisition, Conceptualization. **Martín López-García:** Writing – review & editing, Funding acquisition, Conceptualization. **De Beule Pieter A. A.:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Methodology, Funding acquisition, Conceptualization. **Carlos Calaza:** Writing – review & editing, Methodology. **Ana I. Gómez-Varela:** Writing – review & editing, Conceptualization. **Susana Costa:** Writing – review & editing, Validation. **Hedieh Mahmoodnia:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation. **Rafael Vilarinho:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Diogo Poeta:** Writing – review & editing, Investigation. **Christian Maibohm:** Writing – review & editing, Validation, Methodology, Investigation.

Disclosures

Disclaimer: Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union. Neither the European Union nor the granting authority can be held responsible for them.

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Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Rafael Vilarinho, Hedieh Mahmoodnia, Martín López-Garcia, Adelaide Miranda and Pieter A. A. De Beule have filed an international patent application (PCT Application Number: PCT/IB2025/057293) covering this material and its use as a VOC sensor, If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 3
Qualitative performance comparison between SIPS, PID and GC-MS: comparing SIPS with commercial solutions PID and GC-MS for the detection of VOCs with $E_{IP} < 11$eV.

	Sensitivity	Selectivity	Response Time	Field deployable	Portable
PID	*****	*	*****	Yes	Yes
GC-MS	***	*****	**	No	No
SIPS	**	****	****	Yes	No

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.eti.2026.104895.

Data Availability

Data will be made available on request.

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